MIL-I-21735A(AS)
31 JANUARY 1969
SUPERSEDING
MIL-I-21735(AER)
10 December 1958

MILITARY SPECIFICATION

INDICATOR, AIRSPEED AND MACH NUMBER, PITOT STATIC, 60 to 650 KNOTS

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The indicator covered by this specification shall indicate airspeed and Mach number with the indicated airspeed range from 60 to 650 knots, Mach range from 0.4 to 2.0 Mach, at altitudes from - 1,000 to 50,000 feet, and as listed on MS28089.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

- SPECIFICATIONS

Military

MIL-D-1000/1	Drawings, Engineering and Associated Lists
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-I-5415	Indicators, Airspeed, Pitot-Static Pressure, General Specification for
MIL-C-14806	Coating, Reflection Reducing, For Instrument Cover Glasses and Lighting Wedges
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for

FSC 6610

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SPECIFICATIONS

Military (Continued)

MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for
MIL-I-81400	Instruments, Aircraft, General Specification for

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-470	Maintainability Program Requirements (for Systems and Equipments)
MIL-STD-471	Maintainability Demonstration
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-785	Requirements for Reliability Program (for Systems and Equipments)
MS24367	Lamps, Incandescent - Miniature Integral Lighting
MS24515	Lamp, Subminiature
MS28089	Indicator, Airspeed and Mach Number, Pitot Static, 60 to 650 Knots
MS28105	Window, Dial-Aircraft Instrument Cover Glass
MS33558	Numeral and Letter, Aircraft Instrument Dial, Standard Form of
MS33737	Nut, Self Locking

2.2 Availability of Documents - When requesting any of the applicable documents, refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Publications and Forms Center, Code 1051, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

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3. REQUIREMENTS

3.1 Qualification - The indicator furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable products list.

3.2 General - The requirements specified in MIL-I-5415 are applicable as general requirements of this specification. Where the requirements of MIL-I-5415 and this specification conflict, the requirements of this specification shall govern.

3.2.1 General Design Requirements - Selection and use of parts and materials and general requirements for design and construction shall be in accordance with the requirements of MIL-I-81400 and with the requirements of MIL-I-5415.

3.2.2 Design and Construction - The indicator shall consist of a pitot-static operated indicated airspeed mechanism which will drive a pointer to indicate airspeed on a fixed dial. The indicator shall also contain a static pressure operated altitude mechanism which will drive a moving scale to indicate Mach number. The gearing between the moving scale and the altitude mechanism shall be such that Mach number will be indicated by the pointer on the moving scale at any combination of indicated airspeed and altitude within the range of the instrument.

3.2.2.1 Range - The indicator shall be designed to satisfactorily operate over a range of indicated airspeed of 60 to 650 knots at altitudes from -1,000 to 50,000 feet. The Mach number range shall be from 0.4 to 2.0.

3.2.2.2 Case - Indicator case shall conform to MS28089.

3.2.2.2.1 Body and Mounting Flange - The body and mounting flange of the case shall be made of lightweight nonmagnetic metal, uniform in texture, and shall have a smooth surface. The body shall be finished in a durable black finish and the mounting flange shall be finished with a durable lusterless black finish Color Number 37038 of FED-STD-595. Protective coatings and finishes which will crack, chip, or scale during normal service life or due to extremes of atmospheric conditions shall not be used. The mounting flange shall be capable of retaining clip-in lock nuts conforming to MS33737-11.

3.2.2.2.2 Cover Glass - Cover glass shall be in accordance with MS28105. If the cover glass forms a part of the integral lighting system, then it may deviate from the shape of MS28105 to the extent necessary for integral lighting.

3.2.2.2.3 Anti-reflection Coating - All visible transparent elements shall be coated in accordance with MIL-C-14806 to reduce reflection.

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3.2.2.2.4 Dials - The indicated airspeed scale dial and the Mach scale dial shall conform to Figure 1. Unless otherwise specified, all visible portions of the dials shall be lusterless black Color Number 37038 of FED-STD-595 and all markings on the dial shall be lusterless white Color Number 37875 of FED-STD-595.

3.2.2.2.5 Dial and Cover Glass Location - The distance between the front surface of the bezel and the outside surface of the cover glass shall not exceed 0.062 inch. The distance between the innermost glass surface and the Mach scale dial shall not exceed 0.25 inch.

3 2.2.2.6 Pointer - The pointer shall conform to Figures 1 and 2 and shall be suitable for integral lighting. The pointer shall be firmly attached to the mechanism to prevent slippage and shall be readily removable during overhaul.

3.2.2.2.7 Dials, Pointer and Indices Arrangement - Dials, pointer and indices arrangement shall be as shown in Figure 2.

3.2.2.2.8 Limitation of Indicating Pointer Movement - The pointer movement shall be limited by means of a suitable stop in the mechanism in such a way that the pointer will not be permitted to rotate more than 355 degrees of arc clockwise from the zero graduation when the indicator is submitted to overpressure tests. The design of the stop shall be subject to the approval of the procuring activity.

3.2.2.2.9 Mach Number Setting Index - A triangular Mach number setting index shall be provided as shown in Figure 2. The setting of the Mach number index shall be made by a knob located in the lower left hand corner of the bezel. The knob shall be pushed and turned counterclockwise to set the Mach limit index from 0.4 to 2.0 Mach. The index shall maintain the Mach number setting when the Mach scale dial is rotated pneumatically.

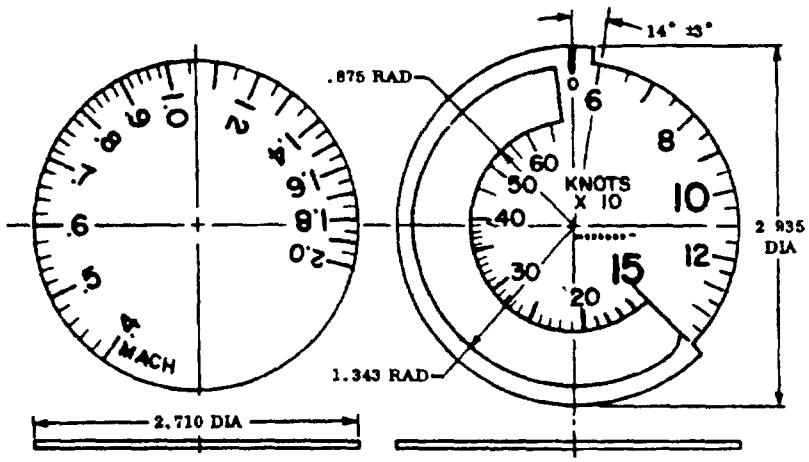
3.2.2.2.10 Airspeed Setting Index - A triangular airspeed setting index shall be provided as shown in Figure 2. The setting of the index shall be controlled by pulling and rotating the knob which controls the Mach number index setting. Setting of either this index or the Mach number index shall not in any way affect the setting of the other. The index shall be settable over a range corresponding to 60 to 145 knots.

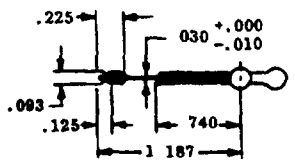
3.3 Weight - The weight of each indicator shall not exceed 1.5 pounds.

3.4 Performance - The indicator shall perform satisfactorily when subjected to the tests specified in Section 4 of this specification.

3.4.1 Reliability Program - The contractor shall establish a reliability program in accordance with MIL-STD-785.

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POINTER

ALL GRADUATIONS ARE PARTIAL RADIAL LINES
DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED.
TOLERANCES DECIMALS ±.010

Marking	Height or Length ± 010	Width of Line ± 005	Finish
Numerals on indicated airspeed scale dial:			
0	.094	016	
6, 8, 12	.188	.031	
20, 30, 40, 50, 60	.156	016	
10, 15	.250	031	
Graduations on indicated airspeed scale dial corresponding to			
5 knot increments	.156	.016	
10 knot increments 60 to 140 inclusive	.188	.032	
10 knot increments 160 to 400 inclusive	.125	.016	
50 knot increments 200 to 650 inclusive	.156	.032	Lusterless
150 knots	625	.032	White
0	.125	.032	Color
Numerals on Mach scale dial	.188	.016	No. 37875 of
Graduations on Mach scale dial corresponding to			FED-STD-595
02 increments	125	.016	
05 increments	156	016	
Numerals 4 to 1 8 inclusive	.188	.032	
Numeral 2.0	156	032	
1.1, 1.3, 1.5, 1.7, and 1.9	188	032	
Lettering "MACH" and "KNOTS X10"	.125	.016	
Mach index and airspeed index (shown on MS28089)	--	--	
Shaded portions of pointer	--	--	
Unshaded portions of pointer	--	--	Lusterless
Background of dials	--	--	Black
Lettering "MS28089-1 or -2" (as applicable)	.047	.010	Color No. 37038 of FED-STD-595

Figure 1. Dial and Pointer Markings

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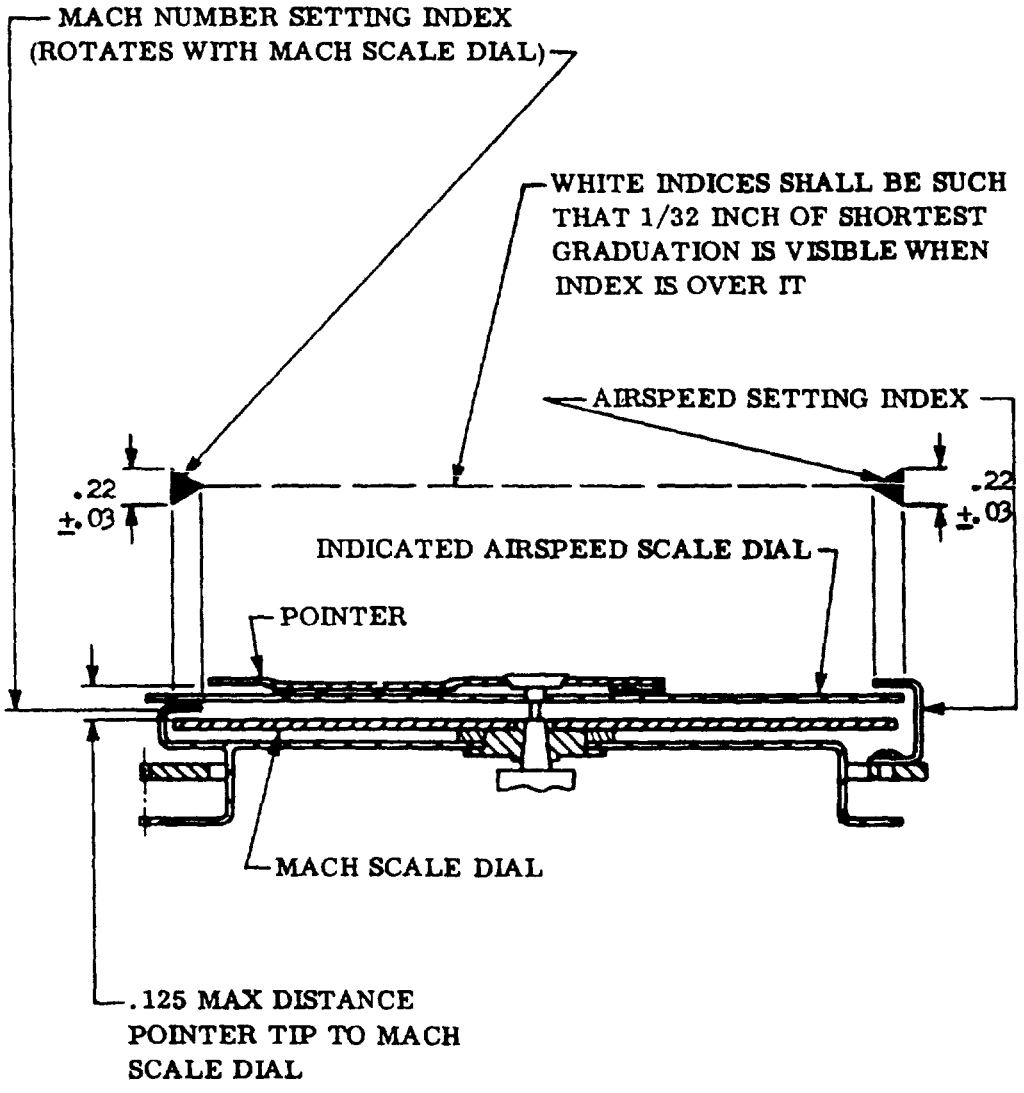


Figure 2. Cross-Section of Dial and Pointer Arrangement with Case Removed

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3.4.1.1 Operational Stability - The indicator shall operate with satisfactory performance, continuously or intermittently, for a period of at least 2000 hours without the necessity for readjustment or relubrication.

3.4.2 Longevity - The indicator shall have a least operating life of 6000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.4.3 Reliability in Mean Time Between Failure (MTBF) - The indicator shall have 2000 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.7.1.

3.5 Electrical Requirements (Integrally Lighted Indicators Only) -

3.5.1 Integral Lighting - The integral lighted indicator, operating on 5 ±0.1 volts AC or DC 5 watts max. for each color using lamps MS24367-715AS15 or MS24515-718AS15, shall illuminate all white markings of the indicator in conformance to MIL-L-25467 and MIL-L-27160.

3.5.2 Electrical Connector - The electrical connector shall be as specified and connected as shown on MS28089.

3.6 Manufacturer's Part Number - Changes in manufacturer's numbers shall be governed by the drawing number requirements of MIL-D-1000/1.

3.7 Maintainability - The contractor's maintainability program shall be in accordance with MIL-STD-470.

3.8 Marking - All markings shall be durable to withstand usage encountered in service. The form of the letters and numerals shall conform to MS33558. Numerals shall distinctly indicate the graduation to which each applies as shown in Figure 1.

3.8.1 Visibility of Dials - Visibility of dials shall be as specified in MIL-I-81400. The integral lamps shall not be visible within the frustum of the cone.

3.8.2 Case Connections - The back of the case adjacent to the applicable connection shall be marked with the letter "P" for the pitot connection and with the word "STATIC" printed twice, forming a ring around the static connection conforming to MS28089. The following shall also be marked on the back of the case

"CAUTION"
"ONLY 5 VOLTS" (for integral lighting only)
"USE STRAIGHT THREAD FITTINGS ONLY"
"DO NOT BLOW IN TUBES".

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3.9 Nameplate - Nameplate, as specified in MIL-I-5415, shall be marked.

INDICATOR, AIRSPEED AND MACH NUMBER
PITOT-STATIC, 60 TO 650 KNOTS
MS28089- (add proper dash number)
MANUFACTURER'S NAME OR TRADEMARK AND SUPPLY CODE
MANUFACTURER'S PART NO. _____
MANUFACTURER'S SERIAL NO. _____
CONTRACT OR ORDER NO. _____
U.S. PROPERTY

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Inspection - Inspection of the indicator shall be classified as follows:

(a) Qualification inspection Qualification inspection consists of examinations and tests performed on sample indicators submitted for approval as a qualified product.

(b) Quality conformance inspection Quality conformance inspection consists of examinations and tests performed on indicators manufactured and submitted for acceptance under contract.

4.3 Qualification Inspection - The qualification inspection of the indicators shall consist of all of the examinations and tests of this specification performed in the order specified under the paragraph headed Inspection Methods and shall include the Reliability Qualification Phase Tests.

4.3.1 Qualification Inspection Sample - Qualification inspection samples shall consist of four indicators manufactured in accordance with this specification. The indicators submitted for qualification inspection shall have been previously subjected only to the individual and sampling plan A inspections. Two samples shall be forwarded with supplier's quantitative test data at the supplier's expense to the laboratory designated in the letter of authorization. (See 6.4.) Two samples shall be

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retained by the supplier for the Reliability Qualification Phase testing which will be run by the supplier at his plant.

4.3.1 1 Sample Identification - Each inspection sample shall be plainly identified by an attached durable tag marked with the following information·

INDICATOR, AIRSPEED AND MACH NUMBER,
PITOT STATIC, 60 TO 650 KNOTS, MS28089 (add proper dash no.)
Submitted (date) by (Manufacturer's name and part Number)
for Sampling Plan B Tests of MIL-I-21735A Authorized by
Contract/order No. _____ or (reference letter authorizing tests).

4.4 Quality Conformance Inspection - The supplier shall furnish all samples and shall be responsible for accomplishing all the inspections designated by the procuring activity. Quality conformance inspection shall be under the supervision of the government quality control representative. The supplier shall furnish test reports showing detailed quantitative results for all tests required by this specification, signed by an authorized representative of the supplier or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product. Quality Conformance Inspection shall consist of the following tests of MIL-I-5415 and as specified herein·

(a) Individual Tests
(b) Sampling Plan A Tests
(c) Sampling Plan B Tests
(d) Reliability Assurance Tests
(e) Longevity Tests
(f) Maintainability Demonstration

4.4.1 Individual Tests - Each indicator submitted for acceptance shall be subjected to and meet the Individual Test. This test shall determine compliance with the requirements of material, workmanship and operational accuracy.

Examination of Product
Leakage
Scale Error at Room Temperature
Hysteresis - After Effect
Friction Error
Position Error
Pointer Zero Position Error
Lighting - Individual (Lighted Indicators Only)

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4.4.2 Sampling Plans - The Sampling Plans shall consist of Sampling Plan A and Sampling Plan B tests. The tests samples selected for sampling tests shall first have passed the Individual test. The test samples which have been subjected to Sampling A or Reliability test shall not be delivered on contract until they have been refurbished and resubmitted and passed all the Individual tests. Test samples which have been subjected to the Sampling Plan B or Longevity test shall not be delivered on contract.

Quantity Offered for Acceptance	Quantity to be Selected for Sampling Plan A Tests
First 15	1 (Zero when Sampling Plan B is invoked)
Next 50	1
Next 75	1
Next 100	1
Each additional 200 or fraction thereof	1

When a defective indicator is detected, no items from those still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. In addition, when a failure occurs, shift to one sample out of the next fifteen and proceed as indicated. (One from the next 50, etc.)

4.4.2.1 Scope of Sampling Plan A Test - Each sample selected for Sampling Plan A Test shall be subjected to the following tests in the order listed

Weight
Magnetic Effect
Damping
High Temperature Operation
Low Temperature Operation
Vibration Error
Lighting Sampling Plan A (lighted indicators only)
Overpressure

4.4.2.2 Sampling Plan B Test - Two indicators shall be selected at random from the first 15 produced on contract and submitted within 10 days after manufacture. These samples shall be forwarded at the supplier's expense to a government laboratory designated by the procuring activity. Each sample shall be plainly identified by a durable tag (see 4.3.1.1).

4.4.2.2.1 Sampling Plan B Approval - Approval of Sampling Plan B indicators shall be by the procuring activity upon satisfactory completion of the designated

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tests. Any design, material or performance defect made evident during this test shall be corrected by the supplier to the satisfaction of the procuring activity. Failure of the sample units to pass any of the tests shall be cause for deliveries of indicators under the contract to cease until proper corrective action is approved and accomplished.

4.4.2.2.2 Scope of Sampling Plan B Test - Each sample selected for Sampling Plan B Test shall be subjected to the following tests in the order listed

Sampling Plan A Tests
Vibration Failure

High Temperature Exposure
Low Temperature Exposure
Acceleration
Humidity
Seasoning
Fungus
Aging
Salt Spray
Lighting Test Sampling Plan B (lighted indicators only)
Mounting Lug
Internal Inspection
Maintainability Demonstration

4.5 Standard Conditions for Test - Unless otherwise specified, all inspection required by this specification shall be made under the following conditions:

Temperature	Room ambient 25° ±5° C
Pressure	Normal atmospheric (approximately 29.92 inches Hg)
Humidity	Room ambient 40 to 90 percent relative humidity
Input Power	5 ±0.1V AC, 400 ±5 cps (Lighted indicators only)

Altitude change rate corresponding to 6,000 feet per minute

4.5.1 Test Readings - Unless otherwise specified, before a test reading is taken the indicator shall be tapped at 60 cps, with a maximum double amplitude of 0.002 inch.

4.5.2 Lighting - The indicator lighting system shall be energized and shall operate satisfactorily during all tests except humidity, seasoning, fungus, aging, mounting lug, and salt spray. The indicator lighting shall operate satisfactorily after completion of each test unless otherwise specified herein.

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4.6 Test Methods - The indicator shall satisfy the performance requirements specified in Section 4 of MIL-I-5415, and as modified herein.

TABLE I

TEST VALUES

Tests and Test Points	Tolerance ±
Friction Error	
Indicator Airspeed Pointer	5.0 knots
Mach Scale	5.0 knots
Position Error:	
Indicator Airspeed Pointer	5.0 knots
Mach Scale	5.0 knots
Pointer Zero Position	1/16 inch
Vibration Error	
Test Altitude - 35,000 feet	
Indicated Airspeed - 350 knots	
Maximum Vibration - Pointer	4.0 knots
Maximum Variation - Pointer	4.0 knots
Overpressure	2.0 knots
Static Connection - 6 inches Water	
Pitot Connection - 40 inches Hg.	
Vibration Failure	2.0 knots
Test Altitude - 35,000 feet	
Indicated Airspeed - 350 knots	
Damping:	
Time required for the airspeed pointer to travel from full scale deflection to the 90 knots graduation	$1.0\ ^{+1.0}_{-0.2}$ seconds
Seasoning	3.0 knots
Aging	3.0 knots

4.6.1 Examination of Product - Examination of Product per MIL-I-5415.

4.6.2 Leakage -

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TABLE II

AIRSPEED SCALE ERROR TEST POINTS AND TOLERANCES

Speed Knots	Pressure Inches of Water	Pressure Inches of Mercury (Hg)	Room Temperature Tolerance Knots	High and Low Temp. Tolerance Knots
60	2.352		2.0	-
70*	3.203		2.0	4.0
80	4.188		2.0	-
90	5.305		2.0	-
100*	6.557		2.0	4.0
110	7.943		2.0	-
120*	9.465		2.0	4.0
130	11.123		2.0	-
140	12.922		2.0	-
150*	14.858		2.5	4.5
160	16.935		2.5	-
170	19.154		2.5	-
180*	21.517		2.5	4.5
190	24.025		2.5	-
200*	26.679		3.0	5.0
220	32.436		3.0	-
250*	42.222		3.0	5.0
280		3.924	3.0	-
300*		4.534	4.0	6.0
320		5.195	4.0	-
350*		6.286	4.0	6.0
380		7.502	4.0	-
400*		8.385	5.0	7.0
450*		10.868	5.0	7.0
500*		13.776	6.0	8.0
550*		17.159	6.0	8.0
600*		21.075	7.0	9.0
650		25.589	7.0	-

Note For sampling and qualification tests use all test points.
For individual tests use test points marked with an asterisk.

4.6.2.1 Case Leak - This test shall be performed as specified in MIL-I-5415.

4.6.2.2 Diaphragm Capsule Leak - This test shall be performed as specified in MIL-I-5415.

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TABLE III

MACH NUMBER SCALE ERROR TEST POINTS AND TOLERANCES

Pressure Altitude		Mach Number	Airspeed Knots	Differential Pressure Inches Hg	Room Temperature Tolerance ± Knots	High and Low Temp. Tolerance ± Knots
Feet X 1000	Inches Hg					
0	29.921	.4	265	3.487	6	—
0*	29.921	.5	331	5.572	6	8
0	29.921	.6	397	8.243	8	—
0	29.921	.7	463	11.583	9	—
0	29.921	.8	529	15.689	10	—
0	29.921	.9	595	20.685	12	—
0*	29.921	1.0	661	26.718	12	17
5	24.896	1.0	613	22.230	12	—
10*	20.577	1.0	566	18.374	11	15
15	16.886	1.0	520	15.078	10	—
20*	13.750	1.0	475	12.278	10	14
25	11.103	1.0	432	9.915	9	—
30*	8.885	1.0	390	7.934	8	11
35	7.041	1.0	350	6.287	7	—
40*	5.538	1.0	313	4.945	6	8
45	4.355	1.0	279	3.889	6	—
50	3.425	.7	165	1.326	4	—
50	3.425	.8	192	1.796	4	—
50	3.425	.9	219	2.367	4	—
50*	3.425	1.0	248	3.058	5	7
50	3.425	1.1	278	3.880	6	—
50	3.425	1.2	309	4.820	6	—
50	3.425	1.4	368	7.018	7	—
50*	3.425	1.6	426	9.606	9	12
50	3.425	1.8	480	12.567	10	—
50	3.425	2.0	532	15.892	11	—

Note: For Sampling and Qualification tests use all test points.
For Individual tests use test points marked with an asterisk.

4.6.3 Scale Error at Room Temperature -

4.6.3.1 Airspeed Scale - This test shall be performed as specified in MIL-I-5415. The errors at the test points shall not exceed those specified in Table II.

4.6.3.2 Mach Number Scale Error - Test shall be conducted by establishing the specified altitude and setting the indicated airspeed pointer on the Mach number

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graduation at each of the specified test points. The error shall then be determined by reading the airspeed value indicated by the airspeed pointer. The errors shall not exceed the tolerances specified in Table III. This test shall be combined with the hysteresis test at each Mach number setting.

4.6.4 Hysteresis - Not more than 15 minutes after the indicator has been subjected to the pressure corresponding to the highest altitude specified in Table III, the pressure shall be increased at a rate corresponding to a decrease in altitude of approximately 3,000 feet per minute until the pressure corresponding to 30,000 feet is reached. The indicator shall remain at this pressure for at least five minutes, but no more than 15 minutes before the test reading is taken. After the reading has been taken, the pressure shall be further increased at the above rate until a pressure corresponding to 10,000 feet is reached. The indicator shall remain at this pressure for at least one minute, but not more than ten minutes before a test reading is taken. After the reading has been taken, the pressure shall be further increased at the same rate until atmospheric pressure is reached. The reading of the Mach scale at either of the two test points shall not differ from the corresponding reading, with decreasing pressure, by more than two knots.

4.6.4.1 Hysteresis After Effect - Not less than one minute and not more than five minutes after the completion of the scale error at room temperature and hysteresis tests, the Mach scale shall be checked to determine that it has returned to its original reading, corrected for any change in atmospheric pressure, within four knots.

4.6.5 Friction Error - Friction error per MIL-I-5415 and Table I shall be conducted for the airspeed pointer and Mach scale dial while it is in normal mounting position.

4.6.6 Position Error - Position error per MIL-I-5415 and Table I, shall be conducted for the airspeed pointer and the Mach scale dial in the following positions; normal, dial face up, dial face down, right side up, right side down, and bottom up.

4.6.7 Pointer Position Error - Pointer position error shall be as specified by MIL-I-5415 and Table I.

4.6.8 Lighting (Individual) - The indicator shall be subjected to and meet the requirements of the individual tests in MIL-L-25467 and MIL-L-27160.

4.6.9 Weight of the indicator shall not exceed 1.5 pounds.

4.6.10 Damping - Damping per MIL-I-5415 and Table I.

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4.6.11 Magnetic Effect - The lighted indicator shall be properly connected and power applied. The indicator shall be tested per MIL-I-5415 by revolving it 360 degrees at the positions 0, 45, 90, 135, 180, 225, 270, and 315 degrees from the initial position Deflection of the compass shall not exceed 1 degree. The test shall be repeated, without power, and the same tolerance shall apply.

4.6.12 Low Temperature Operation (This test may be combined with the Low temperature exposure test at the discretion of the testing agency) - The indicator shall be properly connected except that no pressure shall be applied. The indicator shall then be subjected to an ambient temperature of -54° C (-65° F) for a period of 4 hours. During the last 15 minutes of the 4-hour period, pressure shall be applied to the indicator. At the end of the 4-hour period, and with the temperature maintained at -54° C (-65° F), the indicator shall be tested for scale errors, as specified in the Scale error at room temperature test except that the test points shall be every other point in Tables II and III marked with an asterisk, beginning with the first asterisk. Total allowable scale errors shall not exceed the allowable errors specified in Tables II and III No damage to any part of the indicator shall result from this test.

4.6.13 High Temperature Operation (This test may be combined with the High temperature exposure test at the discretion of the testing agency) - The indicator shall be properly connected except that no pressure shall be applied. The indicator shall then be subjected to an ambient temperature of 71° C (160° F) for a period of 4 hours. During the last 15 minutes of the 4-hour period, power shall be applied to the indicator. At the end of the 4-hour period and with the temperature maintained at 71° C (160° F) the indicator shall be tested for scale errors as specified in the Scale error at room temperature test except that the test points shall be every other point in Tables II and III marked with an asterisk, beginning with the first asterisk. The errors at the test points shall not exceed those specified in Tables II and III. No damage to any part of the indicator shall result from this test.

4.6.14 Vibration Error - The indicator shall be mounted on a vibration stand with a line joining the centers of the two mounting holes in a horizontal plane with the plane of the mounting lugs of the indicator vertical. This test shall be conducted while the test settings specified in Table I are applied. The indicator shall be vibrated with circular motion in a plane inclined 45 degrees to the horizontal plane with a circle diameter between 0.003 and 0.005 of an inch. The frequency of applied vibration shall be varied slowly from 5 to 50 cycles per second. No condition of applied vibration shall cause vibration errors involving pointer or Mach scale vibration or variation in excess of the tolerance specified in Table I.

4.6.15 Lighting (Sampling Plan A) - Lighting (Sampling Plan A) per MIL-L-25467 and MIL-L-27160.

4.6.16 Overpressure - Overpressure per MIL-I-5415 using pressures specified in Table I.

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4.6.17 Vibration Failure - The indicator shall be tested in accordance with Procedure XIII of MIL-E-5272. This test shall be conducted using the test settings specified in Table I. The instrument indication at the test settings specified shall be determined with pressures increasing. At the end of the vibration period the indicator shall be subjected to the pressure specified to produce the test point readings (with pressure increasing). The change in airspeed and/or Mach number indication shall not exceed the tolerance specified in Table I.

4.6.18 Seasoning - Seasoning per MIL-I-5415 at the rate of 3 full scale deflections per minute.

4.6.19 High Temperature Exposure - The indicator shall be tested in accordance with Procedure II of MIL-E-5272 except that the exposure period shall be 24 hours. The indicator shall be returned to room temperature and shall be subjected to and meet all the requirements specified in the individual tests.

4.6.20 Low Temperature Exposure - The indicator shall be tested in accordance with Procedure II of MIL-E-5272. At the conclusion of the 24-hour exposure period return to room temperature. It shall be subjected to and meet all the requirements specified in the individual tests.

4.6.21 Acceleration - Acceleration per MIL-I-5415.

4.6.22 Humidity - The indicator shall be tested per MIL-I-5415.

4.6.23 (Deleted)

4.6.24 Fungus - Fungus per MIL-I-5415.

4.6.25 Aging - Aging per MIL-I-5415.

4.6.26 Salt Spray - The indicator shall be tested per MIL-I-5415.

4.6.27 Lighting (Sampling Plan B) - The lighted type indicator shall be subjected to, and meet the requirements specified in MIL-L-25467 and MIL-L-27160 Sampling Plan B.

4.6.28 Mounting Lug - Mounting lug per MIL-I-5415.

4.6.29 Internal Examination - The case of the indicator shall be opened and the mechanism of the indicator shall be examined. The lamp circuit of the lighted type instrument also shall be checked as specified in MIL-L-25467 and MIL-L-27160. Any deterioration or damage which could in any manner prevent the indicator from meeting functional maintenance or service requirements during service life shall be cause for rejection.

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4.6.30 Maintainability Demonstration - Maintainability Demonstration shall be in accordance with MIL-STD-471 and shall be performed on a minimum of three selected repair tasks. The tasks and timing of the tasks shall be selected in advance and shall be approved by the procuring activity.

4.7 Reliability Assurance Tests - Tests, procedures and reports shall be submitted in accordance with MIL-STD-781. The tests shall be conducted on indicators which have passed the individual tests.

4.7.1 Test Details - Test Level F of MIL-STD-781 shall be used for the 2,000 hour MTBF Qualification and Production Acceptance Tests while cycling each indicator in accordance with the duty cycle and performing the individual tests at 200-hour intervals to indicate that no failure has occurred.

4.7.2 Qualification Phase Tests - Prior to approval of the indicator, between 2 and 6 indicators shall be tested as outlined under "Qualification (Demonstration) Phase of Production Reliability Tests" in MIL-STD-781 using Test Plan III.

4.7 3 Production Acceptance Phase Tests - Prior to acceptance of each lot, between 3 and 5 indicators shall be selected at random by the government inspector from each monthly lot for testing as outlined under "Production Acceptance (Sampling) Phase of Production Reliability Tests" in MIL-STD-781 using Test Plan V.

4.7.4 Longevity Tests - Longevity Tests shall be performed on any two indicators in accordance with Test Plan XXVIII and Test Level F of MIL-STD-781 while cycling each indicator for 6,000 hours per the duty cycle and performing the individual tests at 200-hour intervals to indicate that no failure has occurred. Maintenance may be performed on each indicator provided the total cost of servicing and replacement of parts throughout the longevity test does not exceed the "cost to repair" each indicator as specified in the contract. (See 6.2.)

4.7.5 Duty Cycle - During each hour of duty cycling, the indicator airspeed pointer reading shall be changed from 60 knots to 600 knots and returned to 60 knots within 5 minutes at a rate of 10 exercise cycles per hour with full rotation and return of the Mach scale dial.

4.7.5.1 Performance Characteristics to be Measured - After every 200 hours of duty cycling, each indicator shall meet the Individual Tests. Upon completion of the Reliability Qualification and Production Acceptance Phase Tests, each indicator shall be subjected to and meet all the Individual and Sampling Plan A Tests.

4.8 Reconditioning of Tested Instruments - Reconditioning of tested instruments shall be by the contractor by replacing all wear or damaged items and made "good as new". After reworking the contractor shall resubmit the equipment for acceptance.

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4.9 Rejection and Retest - Indicators which have been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 The packaging, packing and marking requirements for this indicator shall be as specified in Section 5 of MIL-I-5415.

6. NOTES

6.1 Intended Use - The indicator covered by this specification is intended for use in aircraft to give an indication of airspeed and Mach number on a single dial.

6.2 Ordering Data - Procurement documents should specify the following

(a) Title, number and date of this specification.

(b) The quantity, MS part number and type number, if any, of the instrument desired.

(c) Levels of packaging and packing desired.

(d) The laboratory that shall conduct tests.

(e) Reliability test phases (See 4.7).
(f) Cost to repair one indicator (see 4.7.4)

6.3 Definitions - See MIL-I-5415 and MIL-I-81400.

6.4 Qualification - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for the opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. Suppliers are directed to provisions governing qualification (SD-6) and are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command, Department of the Navy, Washington, D.C. 20360, and information pertaining to qualification of products may be obtained from that activity.

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6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence applies.

(a) Contract - The contract shall have precedence over any specification.

(b) This specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(c) Referenced specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.6 Asterisks - Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Philadelphia, Pennsylvania 19112

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119 R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*

SPECIFICATION	
MIL-I-21735A(AS)	INDICATOR, AIRSPEED AND MACH NUMBER, INTEGRALLY LIGHTED PITOT STATIC, 60 to 650 KNOTS

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED *(Printed or typed name and activity)*	DATE

DD FORM 1 NOV 64 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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